

Cognitive Trapeze

*A Strategic Framework for Collaboration with Prof. Daniel Spikol (UCPH) and
Prof. Marcelo Worsley (Northwestern)*

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*Purpose: To present the project, expectations, and capacity-building needs for the 24-month MSCA
fellowship*

July 27, 2026 — updated July 28, 2026

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1. Introduction — Why I Am Writing This Document

I am writing this document to give a clear and honest overview of the MSCA project I am preparing, and to explain how I envision our collaboration. Transparency and mutual understanding from the start are the keys to a successful fellowship.

Marcelo Worsley has kindly recommended you as a potential host supervisor, and I am very excited about this possibility. I have studied your work in multimodal learning analytics, and I believe your expertise is a perfect match for the empirical and analytical core of my project.

2. Project Overview — Cognitive Trapeze

Project Title: Cognitive Trapeze: Modeling Language–Thought Co-Development in AI-Augmented Learning

Duration: 24 months (M1–M24)

Host Institution: University of Copenhagen (Denmark) — under Prof. Spikol's supervision

Co-Supervisor: Prof. Marcelo Worsley (Northwestern University) — Secondment M10–M16

Brief Summary:

The project investigates how language and thought co-evolve in AI-augmented reading environments, using multimodal data and computational modeling. It aims to:

- Formalize the Cognitive Trapeze Model (Bayesian networks, agent-based simulation) — WP1.
- Validate it empirically with 160 learners in 4 Cameroonian schools (ZEP) — WP2.
- Develop a mobile offline AI tutor (ASR + LLM) — WP3.
- Build a sustainable Africa–Europe consortium and an ERC Synergy pre-proposal — WP4.

Key outputs: 4 rank-A articles, an open dataset, a prototype APK, an open-source code repository, and a consortium agreement.

3. Current Status & Planned Pre-print Results

Before the start of the MSCA fellowship, I have already conducted a full ASR evaluation campaign using three Whisper models (Tiny, Medium, Large) across 18 corpus texts, with over 60 learners in real classroom conditions in Cameroon.

The results validate the choice of quantized Whisper Large and clearly demonstrate the need for a semantic correction layer (LLM).

A pre-print summarizing these empirical findings is currently in preparation and will be co-authored with Prof. Marcelo Worsley (Northwestern University). It will be submitted to arXiv immediately following the finalization of the consolidated 18-text analysis. This upcoming preprint will serve as the technical bedrock for the final articles we will co-write during the WP1 and WP3 phases of the project.

4. My Role, Your Role, and Capacity Building per Work Package

The effort estimates below are a starting point for our conversation, not a fixed proposal – I look forward to discussing and adjusting them together based on your availability and priorities.

WP1 – Theoretical Formalization (M1–M8)

Objective: Formalize the Cognitive Trapeze model into Bayesian networks and an agent-based simulation (ABM).

Deliverable: Technical specifications + Article 1 (theoretical, submitted M8).

My Tasks: Literature review and framework specification; design of the Bayesian network; prototyping the ABM in Python (Mesa/NetLogo); writing the full theoretical article.

Support Expected: - From you: Methodological guidance on Bayesian network structure, critical feedback on article drafts, access to UCPH's computational resources. - From Marcelo: Help linking the model to ASR data, review of conceptual sections.

Skills to Strengthen: Bayesian network theory (online courses); Python for ABM (Mesa); academic writing in English.

Contributor	Effort %
Myself	80%
Prof. Spikol	15%
Prof. Worsley	5%

WP2 – Empirical Validation (M6–M22) – Two Field Missions

Phase 2a – First Field Mission (M8–M10): Baseline Assessment (N=160)

My Tasks: Finalize ethics protocol, organize logistics, conduct tests, clean and structure data.

Support Expected: Protocol validation from you; ASR data preparation feedback from Marcelo.

Contributor	Effort %
Myself	90%
Prof. Spikol	5%
Prof. Worsley	5%

Phase 2b – Second Field Mission (M16–M20): MVP Deployment and Pilot Test

My Tasks: Install tablets and train teachers, observe pedagogical fidelity, collect quantitative and qualitative data.

Support Expected: Qualitative analysis supervision from you; technical APK support from Marcelo.

Contributor	Effort %
Myself	85%

Contributor	Effort %
Prof. Spikol	10%
Prof. Worsley	5%

Phase 2c – Final Analysis and Articles (M20–M22)

My Tasks: Advanced data cleaning, Bayesian regression and multilevel models, interpret results and write articles.

Support Expected: Major statistical analysis and co-writing role from you; ASR/LLM review from Marcelo.

Contributor	Effort %
Myself	60%
Prof. Spikol	30%
Prof. Worsley	10%

WP3 – AI Prototyping and Computational Modeling (M10–M20) – Secondment at Northwestern

Objective: Develop the Android APK (quantized Whisper Large + Qwen 0.5B LLM), progression API, and multi-agent simulation.

Deliverable: Beta APK + Article 4 (technical, M22).

My Tasks: Functional specifications, test ASR/LLM models, write technical documentation, co-write Article 4.

Support Expected: Direct technical supervision and architecture validation from Marcelo; multimodal data integration advice from you.

Contributor	Effort %
Myself	40% (specs, testing, writing)
Northwestern Engineers (under Prof. Worsley)	40% (coding)
Prof. Worsley	15% (supervision)
Prof. Spikol	5% (consultation)

WP4 – Consortium Building and Sustainability (M15–M24)

Objective: Formalize a sustainable Africa–Europe–USA partnership and prepare an ERC Synergy pre-proposal.

My Tasks: Formalize institutional agreements between University of Copenhagen (host), the University of Ngaoundéré, and the MINESEC Regional Delegation; organize capacity-building workshop (M18–M20); identify partners and draft ERC pre-proposal (M20–M23); organize final dissemination workshop (M24).

Support Expected: Major role for you — help me navigate the European funding landscape (ERC, UCPH networks), introduce me to potential partners (e.g., other European institutions, relevant research centers), and review the ERC proposal; Marcelo to review technical sections and provide US/EU network access.

Contributor	Effort %
Myself	70%
Prof. Spikol	20%
Prof. Worsley	10%

5. My Personal Capacity Building Roadmap

To be fully effective, I need to strengthen certain skills before and during the project. Here is my priority plan:

Skill Domain	Goal	How to Achieve It	Timeline
Academic English	Write and present confidently	Intensive course (6 months), write abstracts daily, attend seminars	M1–M6
Bayesian modeling	Design and validate BN	Online courses + practice with guidance	M1–M4
Agent-based simulation	Build working ABM in Python	Tutorials + hands-on project with students	M5–M8
Advanced data analysis	Bayesian regression, multilevel models	Workshop (2 weeks) + practice	M10–M12
Android/ML understanding	Understand APK & LLM quantization	Work with engineers, Hugging Face tutorials	M10–M14
ERC proposal writing	Draft competitive pre-proposal	Webinars + mentoring	M15–M22
Scientific writing	Publish in top-rank journals	Co-write articles with supervisors	Continuous

6. Concluding Remarks

I am fully aware that the language of our work will be English. This is my main challenge, but I am absolutely committed to overcoming it. I will take intensive English training from M1, and I will rely on written communication and online tools during the first months.

I am convinced that your supervision, combined with Marcelo's technical expertise, will give me the best possible environment to grow as a researcher and to complete this project with high

impact.

I would be very grateful if we could have a short call to discuss this framework, adjust the expectations, and agree on a shared roadmap.

I look forward to working with you.

Yours sincerely,

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